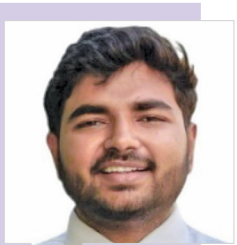


EXPANDING DATA CAPACITY: Merging SDM And WDM Technologies

The integration of multicore fibre and multiband optical networks could redefine data transmission, meeting our growing appetite for speed and capacity.



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With over 65% of the global population now connected to the internet, data demand has surged to unprecedented levels. This relentless growth is fuelled by high-definition video streaming, extensive cloud computing, IoT, high-resolution multiplayer gaming, real-time online transactions, VR, and AR. Data transfer relies on three primary mediums: optical cables, copper cables, and wireless. While copper and wireless media suit short- to medium-range networks, optical fibre networks span thousands of miles, connecting cities, countries, and continents. The rapid expansion of internet users has pushed existing optical fibre infrastructure to its limits, necessitating the adoption of advanced technologies to meet future data needs.

Optical fibre: An overview

An optical fibre carries information signals as light pulses in the near-infrared spectrum and consists of multiple layers (Fig. 1):

Core. The central internal layer of an optical fibre through which light propagates. It is typically composed of high-purity glass and has a higher refractive index than the cladding

Cladding. The layer surrounding the core, with a slightly lower refractive index. Its purpose is to confine the propagated light through total internal reflection, ensuring minimal signal loss

Coating. A thin protective layer that shields the fibre from environmental damage, typically made from polymer materials

Strength member. Present in cables containing multiple fibres, such as aramid yarns, which are surrounded by coated fibres to provide tensile strength

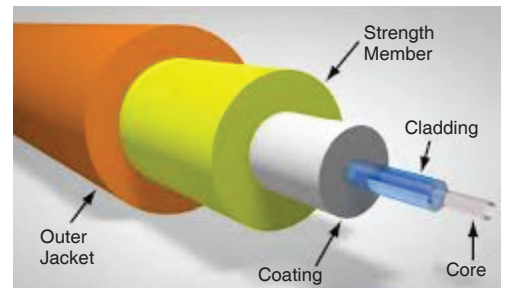


Fig. 1: Cross section of an optical fibre (Source: <https://remee.com>)

Outer jacket. The external layer that safeguards the entire fibre assembly from damage, moisture, and other environmental factors

Based on different modes and refractive index profiles, optical fibres can be categorised into three types (Fig. 2):

Step-index single-mode fibre. A linear refractive index profile from the core to the cladding, supporting a single mode of data transmission

Step-index multi-mode fibre. A linear refractive index profile from the core to the cladding, allowing multiple modes of data transmission

Graded-index multi-mode fibre. A gradually decreasing refractive index profile from the core to the cladding, facilitating multiple modes for data transmission

Current optical fibre technologies are advanced yet approaching their limits. Traditional single-core optical fibres and the most advanced wavelength-division multiplexing (WDM) are also challenged by constraints in bandwidth, fibre thickness, and power due to non-linear effects. According to Nippon Telegraph and Telephone (NTT), Japan's technical review, the maximum transmission